School of Future Tech

Case Study Report

on

CleanEnergyHub - Energy Efficient Appliance Web platform

by

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1. Introduction to the Case Study

This case study focuses on the design and development of a web-based platform called 'CleanEnergyHub'. The platform promotes energy-efficient appliances and helps users understand how energy-efficient products can reduce electricity consumption and lower electricity bills.

The website provides an interactive interface where users can browse energy-efficient products, compare different appliances, calculate their electricity consumption, and contact the store for further assistance. The system is designed using modern web development technologies such as HTML, CSS, and JavaScript.

The objective of this case study is to demonstrate how a simple but practical web application can be developed to educate users about energy savings while also providing an easy platform to explore eco-friendly appliances.

2. Problem Statement / Case Background (Abstract)

Electricity consumption in households is increasing rapidly due to the growing use of electrical appliances. Many consumers are unaware of how much energy their appliances consume or how switching to energy-efficient appliances can reduce their electricity bills.

The goal of this project is to create a website that helps users explore energy-efficient appliances and estimate their electricity consumption. The system also allows users to compare appliances and understand the benefits of choosing products with higher energy efficiency ratings.

The platform includes a product catalog, an appliance comparison system, an electricity consumption calculator, and a contact enquiry form. Together, these features provide users with useful insights into energy consumption and help them make better purchasing decisions.z

3. Case Study Design

- The CleanEnergyHub website is designed as a multi-page web application with the following components:
- Home Page – Introduces the platform and highlights featured energy-efficient appliances.
- Products Page – Displays a list of appliances with filters based on appliance type and star rating.
- Comparison Page – Allows users to compare multiple appliances side-by-side.
- Energy Calculator Page – Calculates electricity consumption and cost based on appliance usage.

- Contact Page – Allows users to submit enquiries related to products or installation.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

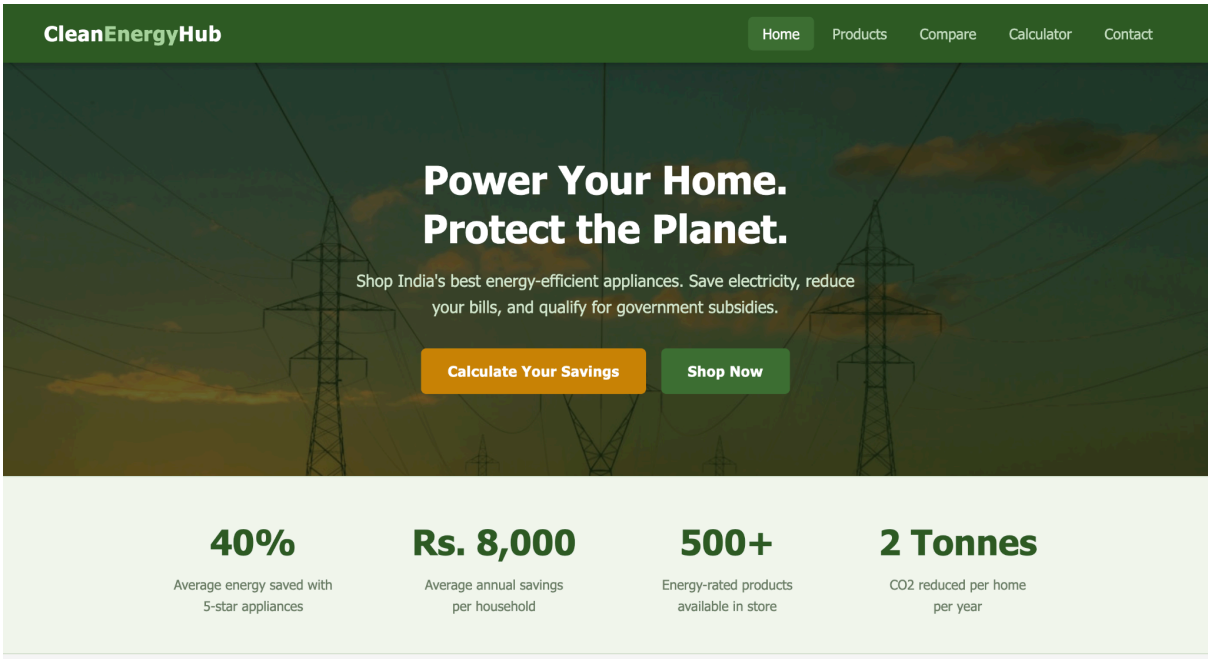
- Technologies used in the development of the project include:
- HTML – Used to create the structure of the web pages.
- CSS – Used to design the layout, colors, and responsiveness of the website.
- JavaScript – Used to implement dynamic functionality such as filtering, calculations, and comparison tables.
- Responsive Web Design – Ensures the website works on both desktop and mobile devices.

5. Problem Statement / Case Study Implementation Details and Snapshots.

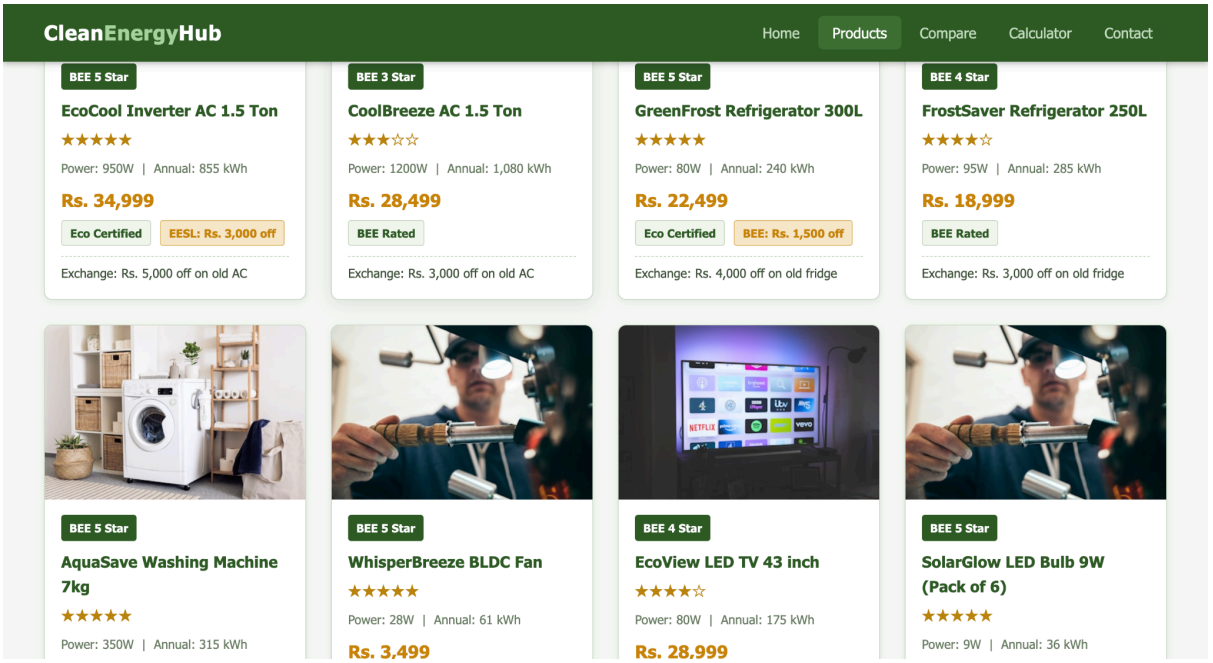
- The implementation of the CleanEnergyHub project includes several modules and interactive components.
- Below are suggested screenshots to include in the report:
- Homepage interface showing featured products.
- Products page with filtering options.
- Appliance comparison table.
- Energy savings calculator interface.
- Contact enquiry form page.

Snapshots:

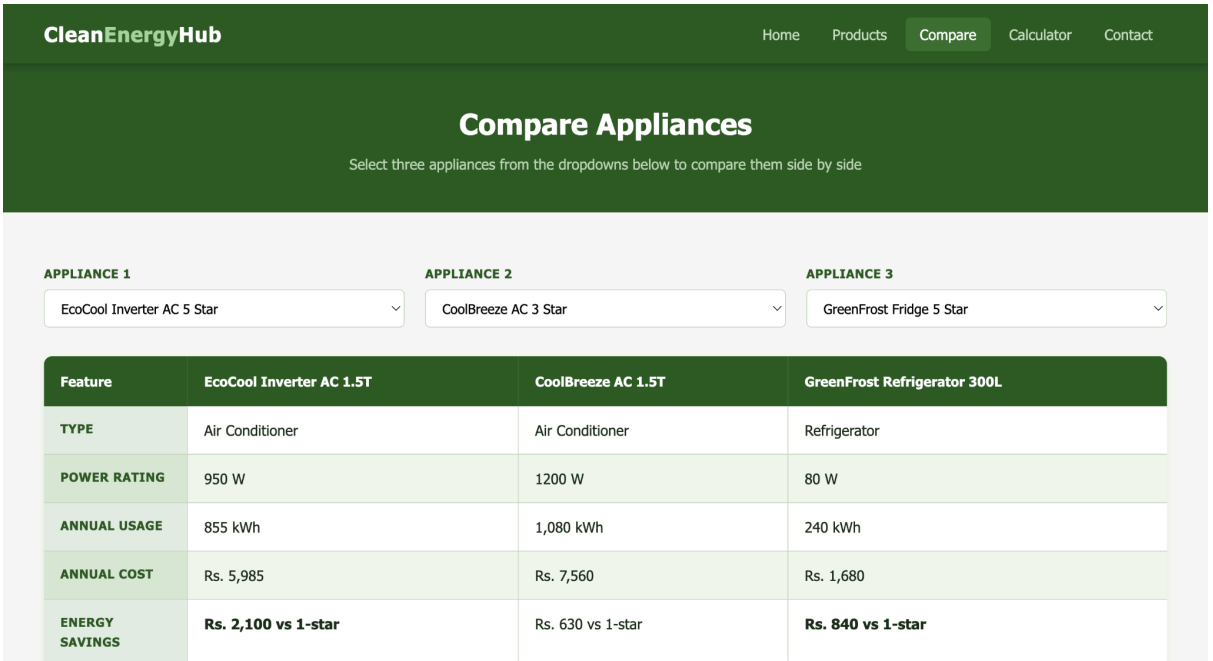
Home Page



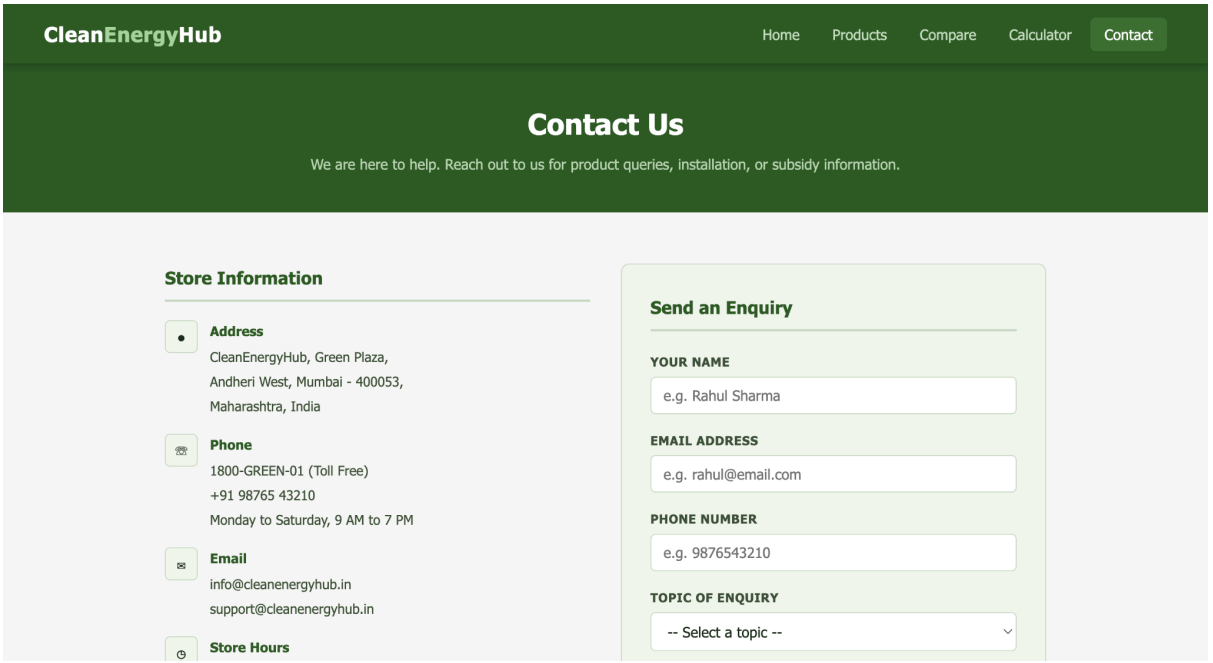
Products Page:



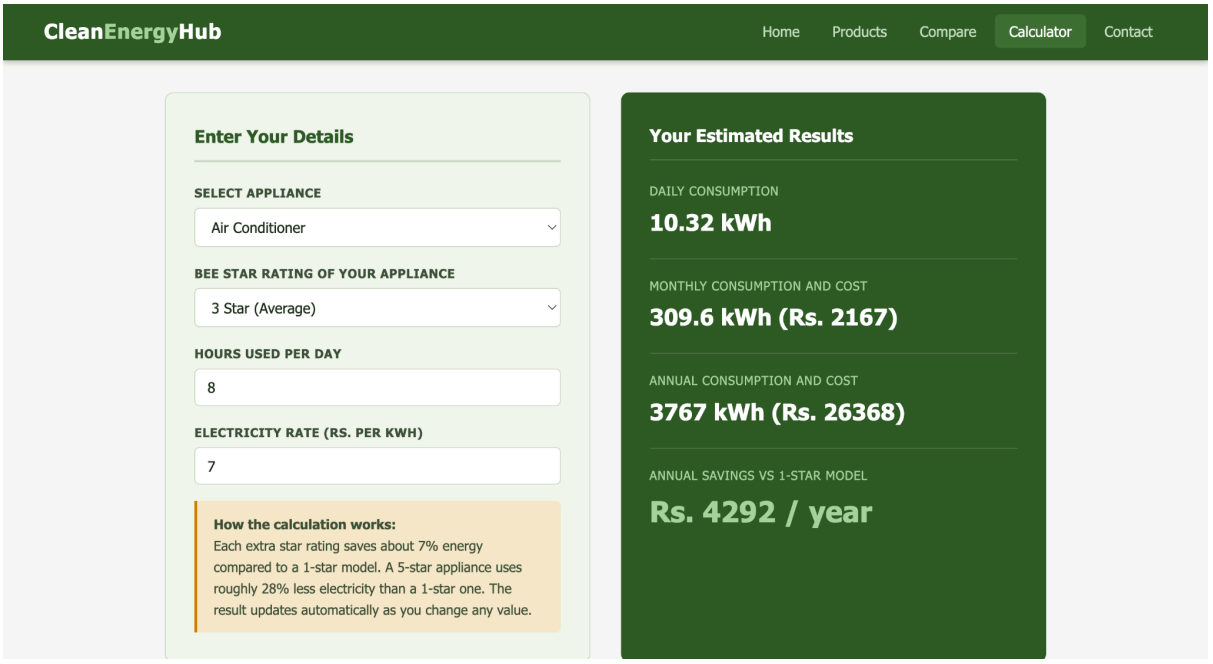
Compare Page :



Contact Us Page :



Calculator Page :



6. Problem Statement / Case Study Results and Conclusion.

The CleanEnergyHub project successfully demonstrates how a simple web application can help users understand energy consumption and make environmentally friendly choices.

The website allows users to explore energy-efficient appliances, compare their features, and calculate their electricity consumption. These features provide valuable insights that can help users reduce their electricity bills.

Overall, the project highlights the importance of energy-efficient appliances and demonstrates how web technologies can be used to create useful and interactive digital solutions.

7. References

- MDN Web Docs – HTML, CSS and JavaScript documentation
- W3Schools – Web development tutorials
- Bureau of Energy Efficiency (BEE) – Energy efficiency standards